

Chapter 2- Planning

Contents:

2.1. Identifying and Selecting Systems Development Projects:

- Introduction;
- Identifying and Selecting Systems Development Projects;
- Corporate and Information Systems Planning

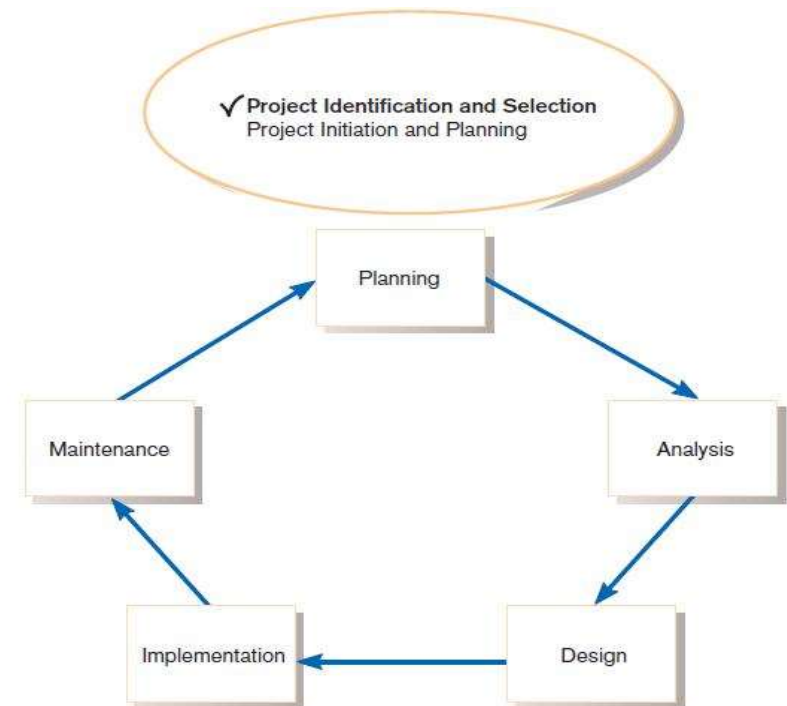
2.2. Initiating and Planning Systems Development Projects:

- Introduction;
- Initiating and Planning Systems Development Projects;
- Process of Initiating and Planning IS Development Projects,
- Assessing Project Feasibility;
- Building and Reviewing the Baseline Project Plan

Identifying and Selecting Systems Development Projects

- **Introduction:**

- The first phase of the SDLC is Identifying and selecting system Development project.
- During this activity, a senior manger, a business group, an IS manager, or a steering committee identify and assess all possible system development projects that an organization unit could undertake.
- Sources of projects
 - Management and business units
 - Managers who want to make a system more efficient or less costly
 - Formal planning groups



The Process of Identifying and Selecting IS Development Projects

- It consists of following activities
 1. Identifying potential development projects
 2. Classifying and ranking projects
 3. Selecting projects for development

The Process of Identifying and Selecting IS Development Projects

1. Identifying potential development projects

- Identification from a stakeholder group.
 - Each stakeholder group brings their own perspective and motivation to the IS decision.
- **Potential development projects are identified by:**
 - **A key member of top management.**
 - CEO of small sized organization
 - Senior executive in a large organization
 - **A steering Committee:** composed of a cross section of manager with an interest in systems
 - **User department**
 - Head of the requesting unit
 - Committee of from the requesting department
 - **The development group or a senior IS manger**

- **Top-management**

- Greater strategic focus
- Largest project size/duration

- **Steering Committee**

- Cross-functional focus
- Greater organizational change
- Larger and riskier project

- **User department**

- Faster development
- Fewer user

- **Development group**

- Fewer development delays

- Top-Down Identification

- Senior management or steering committee
- Focus is on global needs of organization

- Bottom-up Identification

- Business unit or IS group
- Don't reflect overall goals of the organization

The process varies substantially across organizations.

TABLE 4-1 Characteristics of Alternative Methods for Making Information Systems Identification and Selection Decisions

Selection Method	Characteristics
Top Management	Greater strategic focus Largest project size Longest project duration Enterprise-wide consideration
Steering Committee	Cross-functional focus Greater organizational change Formal cost-benefit analysis Larger and riskier projects
Functional Area	Narrow, nonstrategic focus Faster development
Development Group	Fewer users, management layers, and business functions involved Integration with existing systems focus Fewer development delays Less concern with cost-benefit analysis

(Source: Based on McKeen, Guimaraes, and Wetherbe, 1994; GAO, 2000).

2. Classifying and ranking projects

- Second major activity in the project identification and selection process focuses on assessing the relative merit of potential project.
- Performed by top management, steering committee, business units of IS development group
- Value chain analysis is often used
 - Method to analyze an organization's activities to determine where value is added and costs are incurred
- Possible Evaluation Criteria when classifying and Ranking Projects

TABLE 4-2 Possible Evaluation Criteria When Classifying and Ranking Projects

Evaluation Criteria	Description
Value Chain Analysis	Extent to which activities add value and costs when developing products and/or services
Strategic Alignment	Extent to which the project is viewed as helping the organization achieve its strategic objectives and long-term goals
Potential Benefits	Extent to which the project is viewed as improving profits, customer service, and so forth, and the duration of these benefits
Resource Availability	Amount and type of resources the project requires and their availability
Project Size/Duration	Number of individuals and the length of time needed to complete the project
Technical Difficulty/Risks	Level of technical difficulty to successfully complete the project within given time and resource constraints

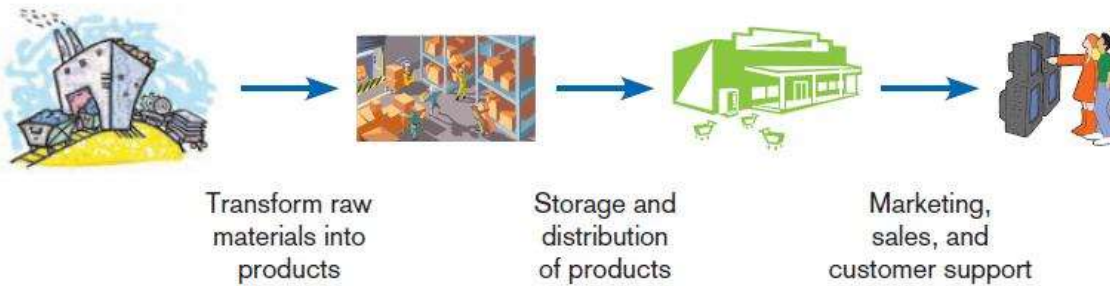


Fig: Organizations can be thought of as a value chain, transforming raw materials into products for customers.

3. *Selecting IS development projects*

- Based on various factors
- Both short- and long-term projects considered
- Most likely to achieve business objectives selected
- Is a very important and ongoing activity

Decision requires consideration of:

- Perceived and real needs
- Potential and ongoing projects
- Current organizational environment
- Existing and available resources
- Evaluation criteria

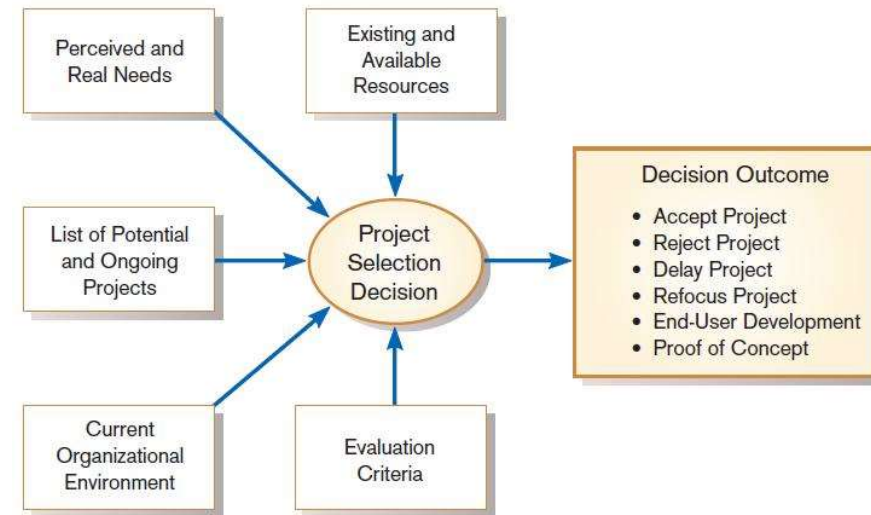


Fig: Project selection decisions must consider numerous factors and can have numerous outcomes.

- One method for deciding among different projects or alternative designs is:
 - For each requirement or constraint:
Score= weight X Rating
 - Each alternative: sum scores across requirements/constraints.
 - Alternative with highest score wins.

Criteria	weight	Alternative A		Alternative B		Alternative C	
		Rating	Score	Rating	Score	Rating	Score
<u>Constraints</u>							
Developer cost	15	4	60	5	75	3	45
Hardware cost	15	4	60	4	60	3	45
Operating cost	5	5	25	1	5	5	25
Total	100		145		140		115

- **Deliverables and Outcomes**
- Primary deliverable from the first part of the planning phase is a **schedule of specific IS development projects**.
- To move to the next part of the planning phase [project initiation and planning](#).
- The outcome of this phase is the assurance that careful consideration was given to project selection and each project can help the organization reach its goals.
- **Incremental commitment:** a strategy in systems analysis and design in which the project is reviewed after each phase and continuation of the project is re-justified.
- **Knowledge of overall organizational business strategy**
 - Improves project selection and identification process
 - Provides sound guidance throughout the systems development life cycle

Corporate and Information Systems Planning

- Traditional Project Identification and Selection

- Solves isolated problems
- Focuses on business processes
- Does not easily allow for organizational change

- Planning-Based Approach to Project Identification and Selection

- Focuses on present and future information needs
- Information needs change slower than business processes

Corporate and Information System Planning

- To benefit from a planning-based approach for identifying and selecting projects, an organization must :
 - Analyze its information needs thoroughly.
 - Plan its projects carefully.

Corporate and Information System Planning

- To benefit from a planning-based approach for identifying and selecting projects, an organization must :
 - Analyze its information needs thoroughly.
 - Plan its projects carefully.
- **Corporate Strategic Planning(CSP)**
 - CSP is an ongoing process that defines the mission, objectives and strategies of an organisation
 - Effective deployment of resources require clearly **understanding of organisational missions, objectives & strategy**
 - Corporate Strategy Planning is a top-down process of three steps
 - CSP is a process that is frequently subject to change
 - It requires constant surveillance

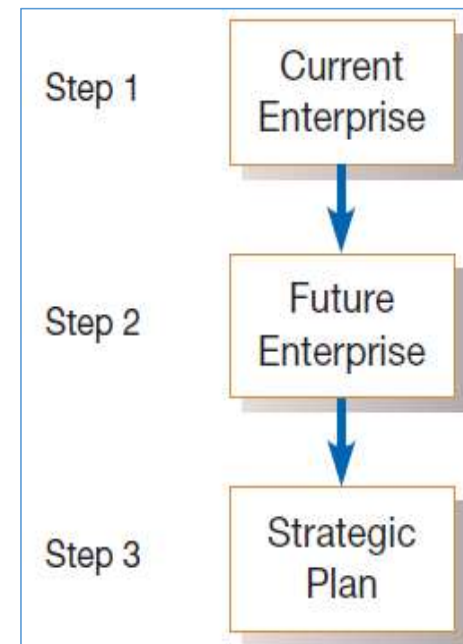


FIGURE 4-6

Corporate strategic planning is a three-step process

Corporate and Information System Planning

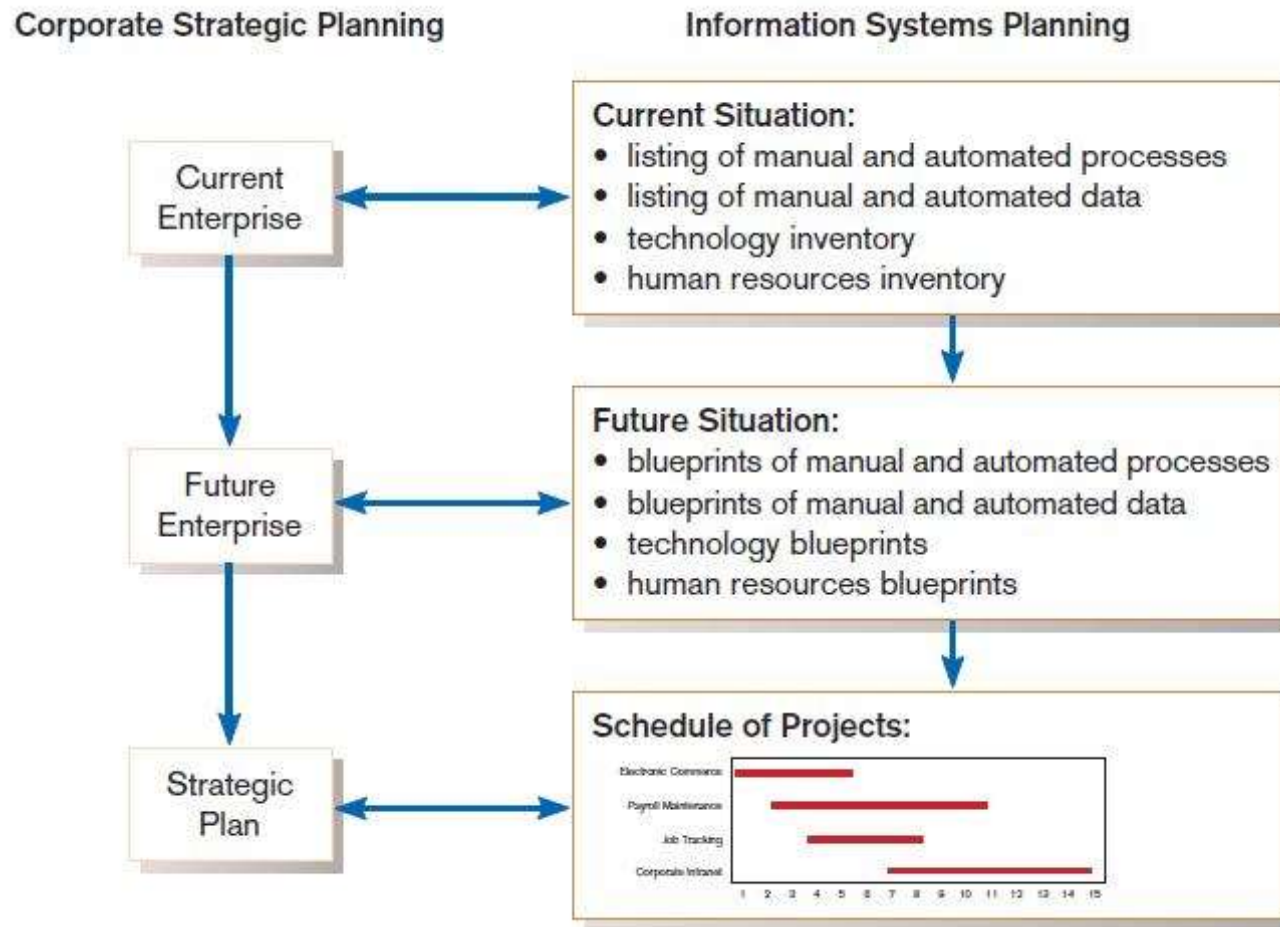


Fig: Parallel activities of corporate strategic planning and information systems planning

- **Corporate Strategic Planning**

- **Mission statement:**

- ☐ A statement that makes it clear what business a company is in.
- ☐ A mission statement is an enduring statement of purpose that distinguishes one business from other similar firms. A mission statement identifies the scope of a firm's operations in product and market terms.
- ☐ A Mission statement describes what the organization is now.

Pine Valley Furniture
Corporate Mission Statement

We are in the business of designing, fabricating, and selling to retail stores high-quality wood furniture for household, office, and institutional use. We value quality in our products and in our relationships with customers and suppliers. We consider our employees our most critical resource.

- **Objective statement:**

- ☐ A series of statements that express an organization's qualitative and quantitative goals for reaching a desired future position
- ☐ "Objectives are end results of planned activity"
- ☐ "Objectives state what is to be accomplished by when and should be quantified if possible."
- ☐ Objectives are always measurable and particular

Pine Valley Furniture
Statement of Objectives

1. PVF will strive to increase market share and profitability (prime objective).
2. PVF will be considered a market leader in customer service.
3. PVF will be innovative in the use of technology to help bring new products to market faster than our competition.
4. PVF will employ the fewest number of the highest-quality people necessary to accomplish our prime objective.
5. PVF will create an environment that values diversity in gender, race, values, and culture among employees, suppliers, and customers.

	Mission Statement	Vision Statement
About	A Mission statement talks about HOW you will get to where you want to be. Defines the purpose and primary objectives related to your customer needs and team values.	A Vision statement outlines WHERE you want to be. Communicates both the purpose and values of your business.
Answer	It answers the question, "What do we do? What makes us different?"	It answers the question, "Where do we aim to be?"
Time	A mission statement talks about the present leading to its future.	A vision statement talks about your future.
Function	It lists the broad goals for which the organization is formed. Its prime function is internal; to define the key measure or measures of the organization's success and its prime audience is the leadership, team and stockholders.	It lists where you see yourself some years from now. It inspires you to give your best. It shapes your understanding of why you are working here.
Change	Your mission statement may change, but it should still tie back to your core values, customer needs and vision.	As your organization evolves, you might feel tempted to change your vision. However, mission or vision statements explain your organization's foundation, so change should be kept to a minimum.
Developing a statement	What do we do today? For whom do we do it? What is the benefit? In other words, Why we do what we do? What, For Whom and Why?	Where do we want to be going forward? Where do we want to reach that stage? How do we want to do it?
Features of an effective statement	Purpose and values of the organization: Who are the organization's primary "clients" (stakeholders)? What are the responsibilities of the organization towards the clients?	Clarity and lack of ambiguity: Describing a bright future (hope); Memorable and engaging expression; realistic aspirations, achievable alignment with organizational values and culture

	Goal	Objective
Meaning	The purpose toward which an endeavor is directed.	Something that one's efforts or actions are intended to attain or accomplish; purpose; target.
Example	I want to achieve success in the field of genetic research and do what no one has ever done.	I want to complete this thesis on genetic research by the end of this month.
Action	Generic action, or better still, an outcome towards which we strive.	Specific action - the objective supports attainment of the associated goal.
Measure	Goals may not be strictly measurable or tangible.	Must be measurable and tangible.
Time frame	Longer term	Mid to short term

Corporate Strategic Planning continued...

Competitive strategy: the method by which an organization attempts to achieve its mission and objectives

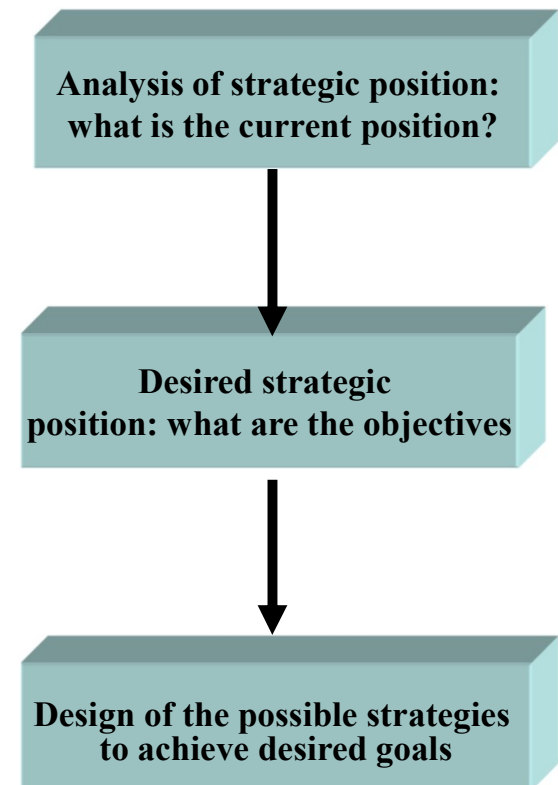
Main types:

- Low-cost producer
- Product differentiation
- Product focus or niche

TABLE 4-3 Generic Competitive Strategies

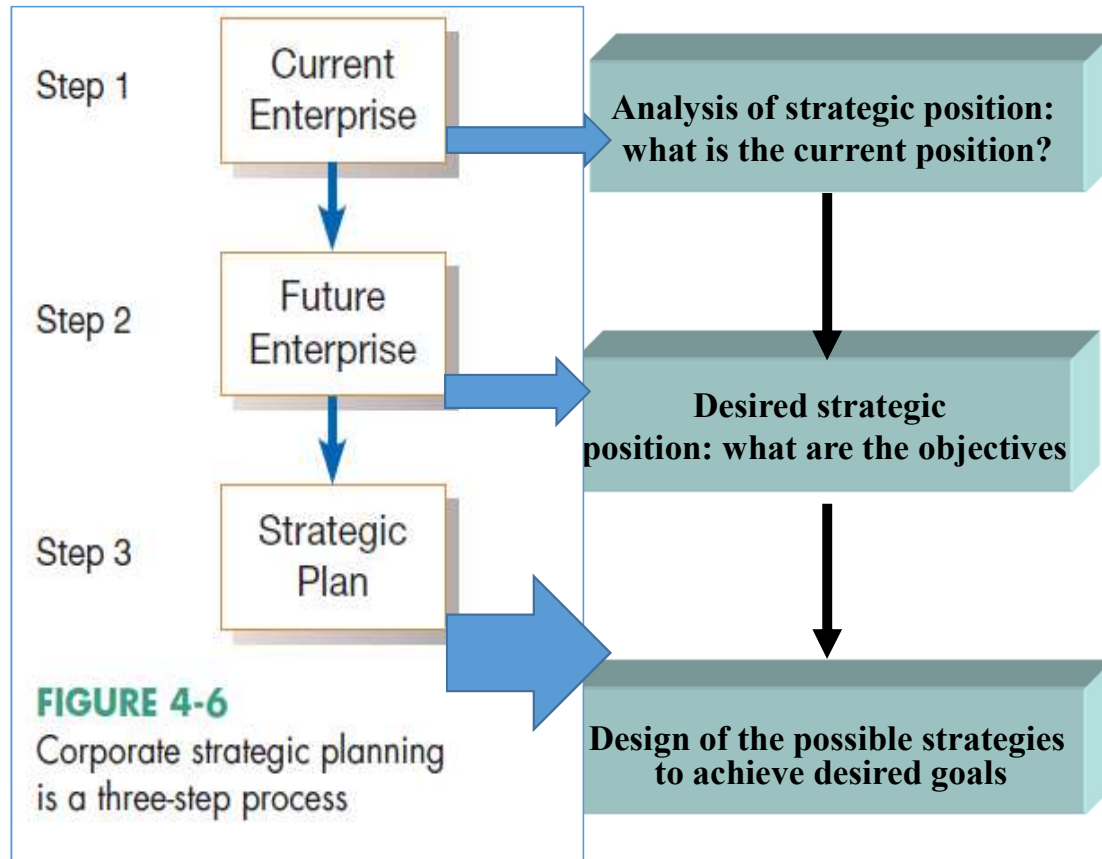
Strategy	Description
Low-Cost Producer	This strategy reflects competing in an industry on the basis of product or service cost to the consumer. For example, in the automobile industry, the South Korean-produced Hyundai is a product line that competes on the basis of low cost.
Product Differentiation	This competitive strategy reflects capitalizing on a key product criterion requested by the market (for example, high quality, style, performance, roominess). In the automobile industry, many manufacturers are trying to differentiate their products on the basis of quality (for example, "At Ford, quality is job one.").
Product Focus or Niche	This strategy is similar to both the low-cost and differentiation strategies but with a much narrower market focus. For example, a niche market in the automobile industry is the convertible sports car market. Within this market, some manufacturers may employ a low-cost strategy while others may employ a differentiation strategy based on performance or style.

Process of Corporate Strategic Planning



Corporate Strategic Planning continued...

Process of Corporate Strategic Planning



- **Information Systems Planning (ISP)**
- Information system planning” is different from “project planning”
- **Information System Planning (ISP)** is the process by which a company (organisation) analyses its information needs and plan its project carefully
- ISP is an **orderly means** of assessing the information needs of an organisation and defining the systems, databases, and technologies that will best satisfy those needs
- ISP is a top-down process that takes into account the outside forces that drive the business (**need vigilance**)
 - Three key activities:
 - Describe the Current Situation
 - Describe the Target (or Future) Situation
 - Develop a Transition Plan and Strategy

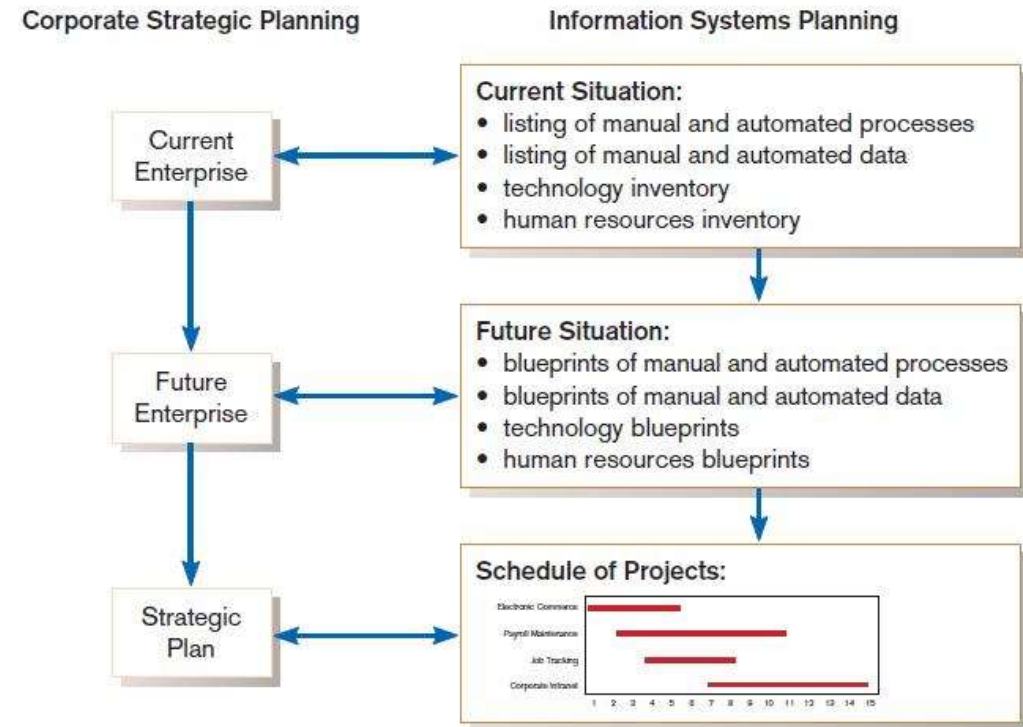


Fig: Parallel activities of corporate strategic planning and information systems planning

Information Systems Planning

1. Describing the Current Situation

- ISP requires support of top management and its commitment to reach desired objectives. Description of the current requires a planning team. **Techniques** used to capture information system needs, include
 - Interviewing managers and executives
 - Reviewing corporate documents
 - Analysing competitors, markets and products
- Consists of Top down and bottom up planning approach
 - Top-down Planning
 - Generic methodology that attempts to gain a broad understanding of the information system needs of the entire organization
 - Bottom-up Planning
 - Generic methodology that identifies and defines IS development projects based upon solving operational business problems or taking advantage of some business opportunities

TABLE 5-4 Advantages to the Top-Down Planning Approach over Other Planning Approaches

Advantage	Description
Broader Perspective	If not viewed from the top, information systems may be implemented without first understanding the business from general management's viewpoint.
Improved Integration	If not viewed from the top, totally new management information systems may be implemented rather than planning how to evolve existing systems.
Improved Management Support	If not viewed from the top, planners may lack sufficient management acceptance of the role of information systems in helping them achieve business objectives.
Better Understanding	If not viewed from the top, planners may lack the understanding necessary to implement information systems across the entire business rather than simply to individual operating units.

- Planning team is chartered to model existing situation
- Identification of Organizational from the information collected from current situations are:
 - **Locations**: where the company operates
 - **Units**: that operate within a company
 - **Functions**: cross organisational collection of activities used to perform day-to-day business operation.
 - **Processes**: of a company
 - **Data**: underlying processes
 - **Information systems**: represent automated and non automated systems used to support business processes.
- Matrices are developed to cross-reference units
 - Location-to-Function
 - Location-to-Unit
 - Unit-to-Function
 - Function-to-Objective
 - Function-to-Process
 - Function-to-Data Entity
 - Process-to-Data Entity
 - Process-to-Information System
 - Data Entity-to-Information System
 - Information System-to-Objective

FUNCTIONS:

- business planning
- product development
- marketing and sales
- production operations
- finance and accounting
- human resources

DATA ENTITIES:

- customer
- product
- vendor
- raw material
- order
- invoice
- equipment

INFORMATION SYSTEMS:

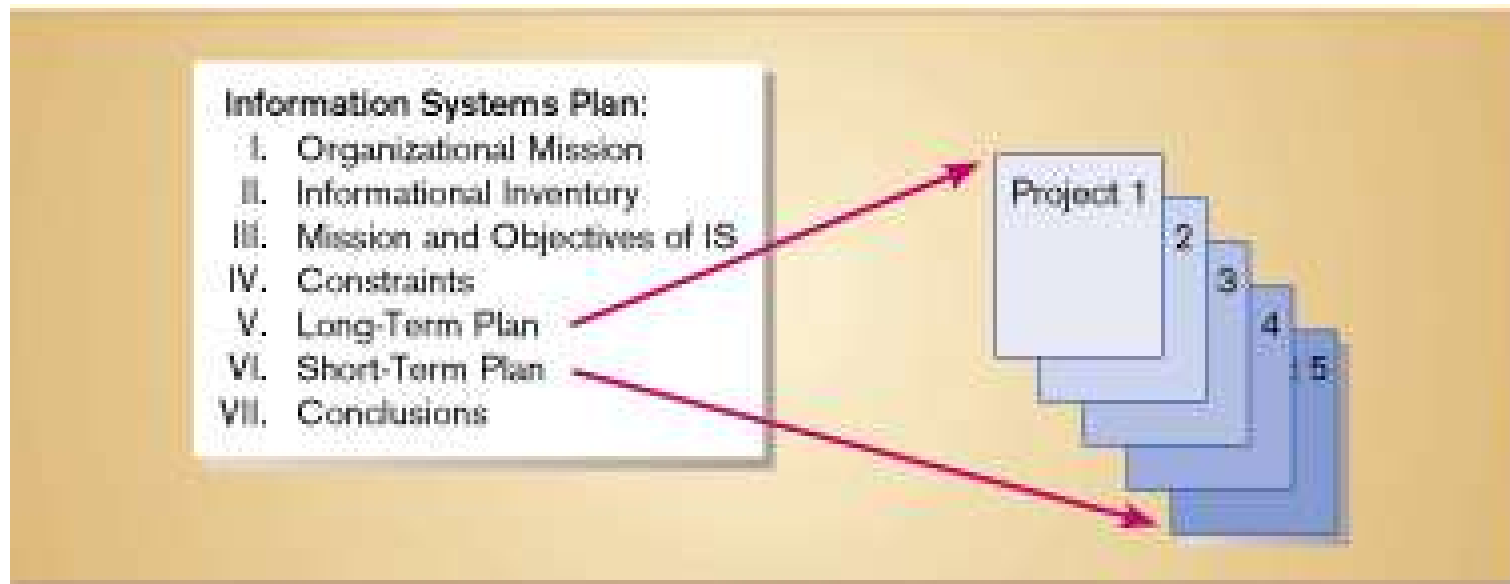
- payroll processing
- accounts payable
- accounts receivable
- time card processing
- inventory management

2. Describing the Target Situation

- Update list of organizational locations, functions, etc. to reflect desired locations, functions, etc.
- Matrices are updated to reflect future states
- Planners focus on differences between current lists and matrices and future lists and matrices

3. Developing a Transition Strategy and Plans

- Broad, comprehensive document that looks at both short and long-term organizational development needs
- Plans include organisational mission, information inventory, mission and objectives of IS, constraints, long-term plans and short-term plans
- Selected projects are those derived from ISP and that fulfil the gap between current and desired situation



Advantages of Information System Planning(ISP)

- Allow to construct IS that support the whole organisation
- Allow integration of different tools
- Avoid data redundancy (process oriented approach vs data oriented approach)
- Reduce maintenance costs because planned system will require less revision

Initiating and Planning Systems Development Projects

- Introduction;
- Initiating and Planning Systems Development Projects;
- Process of Initiating and Planning IS Development Projects,
- Assessing Project Feasibility;
- Building and Reviewing the Baseline Project Plan

Initiating and Planning Systems Development Projects

- **Contemplative Questions**

- What deliverables are produced in early stages of a project?
- What is the scope of this project? What *should the scope be*?
- If I were a **member of the steering committee or an investor**... would I invest in this project?
- If I were a **project manager**... would I stake my reputation on this project? Is it feasible?
- If I were an **analyst**... what is the problem being solved here? Is this *really* the problem, or just a symptom? What is the *real* problem the proposed system is trying to address?
- What must be considered when making the decision on the division between project initiation and planning (PIP) and analysis.
- How much effort should be expended on the PIP process?
- Who is responsible for performing the PIP process?
- Why is PIP such a challenging activity?

The process of Initiating and Planning Systems Development Projects

• **Initiation Vs Planning**

- For any single project...
 - When is it initiated?
 - When does planning start?
- “Formal” Project Initiation
 - A project team is established.
 - The project is given a number for tracking purposes.
 - The project may be given a name.
 - A relationship with a customer may be developed, or an existing relationship may continue.
- “Formal” Project Initiation Plan
 - Procedures are established for management, decision-making, communication.
 - Milestones are identified and possibly negotiated.
- Project initiation focuses on activities designed to assist in organizing a team to conduct project planning.
- *Establishing the Project Initiation Team.*
- *Establishing a Relationship with the Customer*

The process of Initiating and Planning Systems Development Projects

- **Elements of Project Initiation**

- Establishment of project **team**
- Development of **relationship** with customer
- Establishing the project **Initiation Plan**
- Establishment of Management **Procedures**
- Establishment of Project **Workbook** and Project Management Environment
- Developing the **Project Charter**

- The key activity of project initiation is the development of the *project charter*.

- **Project Charter:**

- A short document that is prepared for both internal and external stakeholders.
- Provides a high-level overview of the project.
- Useful communication tool that helps to assure that the organizations and other stakeholders understand the initiation of a project
- A project charter typically contains:
 - Project title and date of authorization
 - Project manager name and contact information
 - Customer name and contact information
 - Projected start and completion dates
 - Key stakeholders, project role, and responsibilities
 - Project objectives and description
 - Key assumptions or approach
 - Signature section for key stakeholders

The process of Initiating and Planning Systems Development Projects

- The key activity of project planning is the process of defining clear, discrete activities and the work needed to complete each activity within a single project.
- The objective of the project planning process is the development of a *Baseline Project Plan (BPP)* and the *Project Scope Statement (PSS)*
- **Project Planning**
- **Is the process of defining clear, discrete activities and the work needed to complete each activity, the elements are:**
 - Describe the project **scope, alternative, and feasibility.**
 - Divide project into manageable **tasks.**
 - Estimate **resources**, and create resource plan.
 - Developing a preliminary **schedule.**
 - Developing a **communication plan.**
 - Determining project **standards and procedures.**
 - Identifying and **assessing risks.**
 - Creating a preliminary **budget.**
 - Developing a **Statement Of Work.**
 - Setting a **Baseline Project Plan.**

The process of Initiating and Planning Systems Development Projects

- Major Deliverables and Outcomes of the plan:
 - **Business Case**
 - Justification for an information system.
 - Presented in terms of the tangible and intangible economic benefits and costs.
 - The technical and organizational feasibility of the proposed system
 - **Project Scope Statement (PSS)**
 - A document prepared for the customer.
 - Describes what the project will deliver.
 - Outlines at a high level all work required to complete the project.
 - **Baseline Project Plan (BPP), contains the best estimates of :**
 - A major outcome and deliverable from the PIP phase.
 - Contains the best estimate of a project's scope, benefits, costs, risks, and resource requirements.
 - **Statement of Work (SOW)**
 - Describes deliverables
 - Outlines work needed to be performed

Figure 5-2

Statement of Work for the Customer Tracking System (Pine Valley Furniture)

Pine Valley Furniture Statement of Work		Prepared: 0/06/05
Project Name:	Customer Tracking Systems	
PVF Project Manager:	Jim Woo	
Customer:	Marketing	
Project Sponsor:	Jackie Judson	
Project Start/End (projected):	10/1/04–2/1/06	
PVF Development Staff Estimates (man-months):		
Programmers:	2.0	
Jr. Analysts:	1.5	
Sr. Analysts:	0.3	
Supervisors:	0.1	
Consultants:	0.0	
Librarian:	0.1	
TOTAL:		4.0
Project Description		
Goal		
This project will implement a customer tracking system for the marketing department. The purpose of this system is to automate the ... to save employee time, reduce errors, have more timely information, ...		
Objective		
• minimize data entry errors • provide more timely information • ...		
Phases of Work		
The following tasks and deliverables reflect the current understanding of the project:		
In Analysis, ...		
In Design, ...		
In Implementation, ...		

Statement of Work (SOW) is a “contract” between the IS staff and the customer regarding deliverables and time estimates for a system development project.

Figure 5-3

System Service Request for Customer Tracking System (Pine Valley Furniture)

Pine Valley Furniture System Service Request	
REQUESTED BY	Jackie Judson
DATE	August 22, 2005
DEPARTMENT	Marketing
LOCATION	Headquarters, 570c
CONTACT	Tel: 4-3290 FAX: 4-3270 e-mail: judson
TYPE OF REQUEST	URGENCY
☒ New System	☐ Immediate -- Operations are impaired or opportunity lost
☐ System Enhancement	☐ Problems exist, but can be worked around
☐ System Error Correction	☒ Business losses can be tolerated until new system installed
PROBLEM STATEMENT	
Sales growth at PVF has caused a greater volume of work for the marketing department. This volume of work has greatly increased the volume and complexity of the data we need to deal with and understand. We are currently using manual methods and a complex PC-based electronic spreadsheet to track and forecast customer buying patterns. This method of analysis has many problems: (1) we are slow to catch buying trends as there is often a week or more delay before data can be taken from point-of-sales system and manually enter it into our spreadsheet; (2) the process of manual data entry is prone to errors (which makes the results of our subsequent analysis suspect); and (3) the volume of data and the complexity of analyses conducted in the system soon to be overwhelming our current system—sometimes the program starts recalculating and never returns while for others it returns information that we know cannot be correct.	
SERVICE REQUEST	
I request a thorough analysis of our current method of tracking and analysis of customer purchasing activity with the intent to design and build a completely new information system. This system should handle all customer purchasing activity, support display and reporting of critical sales information, and assist marketing personnel in understanding the increasingly complex and competitive business environment. I feel that such a system will improve the competitiveness of PVF, particularly in our ability to better serve our customers.	
IS LIAISON	Jim Woo, 4-6207 FAX: 4-6200 e-mail: jwoo
SPONSOR	Jackie Judson, Vice-President, Marketing
TO BE COMPLETED BY SYSTEMS PRIORITY BOARD	
☐ Request approved	Assigned to _____
	Start date _____
☐ Recommend revision	
☐ Suggest user development	
☐ Reject for reason _____	

System Service Request (SSR) is a form requesting development or maintenance of an information system. It includes the contact person, a problem statement, a service request statement, and liaison contact information.

Assessing Project Feasibility

- **Six Categories of feasibility:**

- Economic
- Technical
- Operational
- Schedule
- Legal and contractual
- Political

- **Assessing Economic Feasibility:**

- A process of identifying the financial benefits and costs associated with a development project.
 - Often referred to as *cost-benefit analysis*.
 - Project is reviewed after each SDLC phase in order to decide whether to continue, redirect, or kill a project.
 - **Determine Benefits**
 - **Tangible Benefits**
 - Refer to items that **Can be measured easily**
 - **Examples**
 - Cost reduction and avoidance
 - Error reduction
 - Increased flexibility
 - Increased speed of activity
 - Improved management planning and control
 - Opening new markets and increasing sales opportunities

Intangible Benefits

- are benefits derived from the creation of an information system that cannot be easily measured in dollars or with certainty. **Cannot be measured easily**

Examples

Increased employee morale
Competitive necessity
More timely information
Promotion of organizational learning and understanding

Table 5-3 Intangible Benefits from the Development of an Information System

• Competitive necessity	• Faster decision making
• More timely information	• Information processing efficiency
• Improved organizational planning	• Improved asset utilization
• Increased organizational flexibility	• Improved resource control
• Promotion of organizational learning and understanding	• Increased accuracy in clerical operations
• Availability of new, better, or more information	• Improved work process that can improve employee morale
• Ability to investigate more alternatives	• Positive impacts on society

Accessing Economic Feasibility

- **Determine Costs**

- **Tangible Costs**

- a cost associated with an information system that can be measured in dollars and with certainty.
 - IS development tangible costs include:
 - Hardware costs,
 - Labor costs, or
 - Operational costs including employee training and building renovations.

- **Intangible Costs**

- a cost associated with an information system that cannot be easily measured in terms of dollars or with certainty.
 - Intangible costs can include:
 - Loss of customer goodwill,
 - Employee morale, or
 - Operational inefficiency.

Accessing Economic Feasibility

- **One-Time Costs**

- Associated with project startup, initiation and development
- Includes
 - System Development
 - New hardware and software purchases
 - User training
 - Site preparation
 - Data or system conversion

- **Recurring Costs**

- a cost resulting from the ongoing evolution and use of a system.
- Examples of these costs include:
 - Application software maintenance,
 - Incremental data storage expenses,
 - Incremental communications,
 - New software and hardware leases, and
 - Supplies and other expenses (i.e. paper, forms, data center personnel).
- Both one-time and recurring costs can consist of items that are fixed or variable in nature.
- *Fixed costs* are billed or incurred at a regular interval and usually at a fixed rate.
- *Variable costs* are items that vary in relation to usage

Determining Project Costs

- Possible Information System Costs are :
 - Procurement
 - Consulting, equipment, site preparation, capital, management time
 - Start-up
 - Operating systems, communications installation, personnel hiring, organizational disruption
 - Project-related
 - Application software, software modification, personnel overhead, training, data analysis, documentation
 - Operating
 - System maintenance, rental, asset depreciation, operation and planning

TABLE 5-4 Possible Information Systems Costs

Type of Cost	Examples	Type of Cost	Examples
Procurement	Hardware, software, facilities infrastructure Management and staff Consulting and services	Project	Infrastructure replacement/improvements Project personnel Training Development activities Services and procurement Organizational disruptions Management and staff
Start-Up	Initial operating costs Management and staff Personnel recruiting	Operating	Infrastructure replacement/improvements System maintenance Management and staff User training and support

TABLE 5-5 Guidelines for Better Cost Estimating

1. Have clear guidelines for creating estimates.
2. Use experienced developers and/or project managers for making estimates.
3. Develop a culture where all project participants are responsible for defining accurate estimates.
4. Use historical data to help in establishing better estimates of costs, risks, schedules, and resources.
5. Update estimates as the project progresses.
6. Monitor progress and record discrepancies to improve future estimates.

(Source: Based on Lederer and Prasad, 1992; Hubbard, 2007; Sonje, 2008.)

The Time Value of Money(TVM)

- **Time value of money (TVM):** the concept that money available today is worth more than the same amount tomorrow.
- Three Financial Measurements for Economic Feasibility are:
 - **Net Present Value (NPV)**
 - Use discount rate to determine present value of cash outlays and receipts
 - **Return on Investment (ROI)**
 - Ratio of cash receipts to cash outlays
 - **Break-Even Analysis (BEA)**
 - Amount of time required for cumulative cash flow to equal initial and ongoing investment
- Other Related Terminologies are:
 - **Discount rate:** the rate of return used to compute the present value of future cash flows (*the cost of capital*).
 - **Present value:** the current value of a future cash flow

TABLE 5-6 Commonly Used Economic Cost–Benefit Analysis Techniques

Analysis Technique	Description
Net Present Value (NPV)	NPV uses a discount rate determined from the company's cost of capital to establish the present value of a project. The discount rate is used to determine the present value of both cash receipts and outlays.
Return on Investment (ROI)	ROI is the ratio of the net cash receipts of the project divided by the cash outlays of the project. Trade-off analysis can be made among projects competing for investment by comparing their representative ROI ratios.
Break-Even Analysis (BEA)	BEA finds the amount of time required for the cumulative cash flow from a project to equal its initial and ongoing investment.

Time Value of Money

- **Net Present Value**

- PV_n = *present value* of Y dollars n years from now based on a **discount rate** of i .
- NPV = sum of PVs across years.
- Calculates *time value of money*.

$$PV_n = Y \times \frac{1}{(1 + i)^n}$$

- The rate at which money can be borrowed or invested is referred to as the *cost of capital*, and is called the **discount rate** for TVM calculations

Example

- Suppose you want to buy a used car from an acquaintance and she asks that you make three payments of \$1500 for three years, beginning next year, for a total of \$4500. If she would agree to a single lump-sum payment at the time of sale (and if you had the money!), what amount do you think she would agree to? Should the single payment be \$4500? Should it be more or less?
- **To answer this question, we must consider the time value of money.**
 - Let's suppose that the seller could put the money received for the sale of the car in the bank and receive a **10 percent return on her investment**. A simple formula can be used when figuring out the **present value** of the three \$1500 payments

where PV^n is the present value of Y dollars n years from now when i is the discount rate.

From our example, the present value of the three payments of \$1500 can be calculated as

$$PV_1 = 1500 \times \frac{1}{(1 + .10)^1} = 1500 \times .9091 = 1363.65$$

$$PV_2 = 1500 \times \frac{1}{(1 + .10)^2} = 1500 \times .8264 = 1239.60$$

$$PV_3 = 1500 \times \frac{1}{(1 + .10)^3} = 1500 \times .7513 = 1126.95$$

where PV_1 , PV_2 , and PV_3 reflect the present value of each \$1500 payment in years 1, 2, and 3, respectively.

To calculate the *net present value (NPV)* of the three \$1500 payments, simply add the present values calculated previously

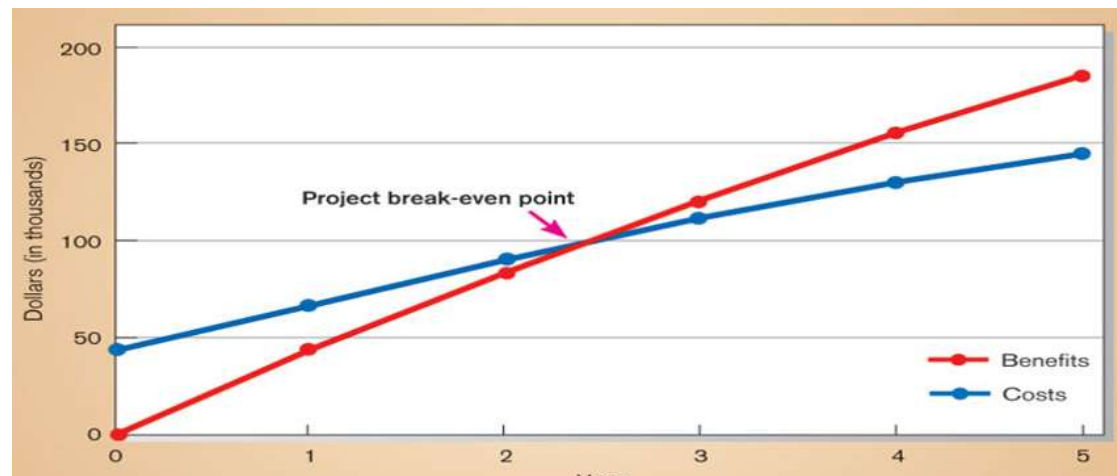
$$\begin{aligned} (NPV &= PV_1 + PV_2 + PV_3 \\ &= 1363.65 + 1239.60 + 1126.95 \\ &= \$3730.20). \end{aligned}$$

In other words, the seller could accept a lump-sum payment of \$3730.20 as equivalent to the three payments of \$1500, given a discount rate of 10 percent.

Break-even analysis

- The objective of the break-even analysis is to discover at what point (if ever) benefits equal costs (i.e., when breakeven occurs).
- To conduct this analysis, the NPV of the yearly cash flows are determined. The yearly cash flows are calculated by subtracting both the one-time cost and the present values of the recurring costs from the present value of the yearly benefits. The overall NPV of the cash flow reflects the total cash flows for all preceding years.

$$\text{Break-Even Ratio} = \frac{\text{Yearly NPV Cash Flow} - \text{Overall NPV Cash Flow}}{\text{Yearly NPV Cash Flow}}$$



Rules:

1. $PV_n = Y \times \frac{1}{(1+i)^n}$
2. $NPV \text{ of all Benefits} = PV_1 \text{ of benefits} + PV_2 \text{ of benefits} + \dots + PV_n \text{ of benefits}$
3. $NPV \text{ of all costs} = \text{One time Costs} + PV_1 \text{ of Recurring Costs} + PV_2 \text{ Recurring Costs} + \dots + PV_n \text{ Recurring Costs}$
4. $\text{Overall NPV} = NPV \text{ of all Benefits} - NPV \text{ of all costs}$
5. $\text{Over all ROI} = \left(\frac{\text{Over all NPV}}{NPV \text{ of all costs}} \right)$
6. $\text{Yearly NPV Cash flow} = PV \text{ of Benefits} - PV \text{ of costs}$
7. $\text{Overall NPV Cash Flow} = NPV \text{ of all Benefits} - NPV \text{ of all costs}$
8. $\text{Break - Even Ratio(fraction)} = \frac{\text{Yearly NPV Cash Flow} - \text{Over all NPV Cash flow}}{\text{Yearly NPV Cash Flow}}$

Economic Feasibility Analysis –Example

- Lets assume that the following Cost are Benefits are Identified for the Customer Tracking Projects (given in the next slide)
 - Calculate NPV
 - Calculate ROI
 - Calculate Break Even Ratio and Break Even Year (Payback Period)

TANGIBLE BENEFITS WORKSHEET
Customer Tracking System Project

Year 1 through 5

A. Cost reduction or avoidance	\$ 4,500
B. Error reduction	2,500
C. Increased flexibility	7,500
D. Increased speed of activity	10,500
E. Improvement in management planning or control	25,000
F. Other _____	0
TOTAL tangible benefits	\$50,000

ONE-TIME COSTS WORKSHEET
Customer Tracking System Project

Year 0

A. Development costs	\$20,000
B. New hardware	15,000
C. New (purchased) software, if any	
1. Packaged applications software	5,000
2. Other _____	0
D. User training	2,500
E. Site preparation	0
F. Other _____	0
TOTAL one-time costs	\$42,500

RECURRING COSTS WORKSHEET
Customer Tracking System Project

Year 1 through 5

A. Application software maintenance	\$25,000
B. Incremental data storage required: 20 GB \$50 (estimated cost/GB = \$50)	1000
C. Incremental communications (lines, messages, . . .)	2000
D. New software or hardware leases	0
E. Supplies	500
F. Other _____	0
TOTAL recurring costs	\$28,500

Economic Feasibility Analysis of "Customer Tracking Project"							
Year of project	0	1	2	3	4	5	(Total in \$)
Net Economic Benefit	0	50000	50000	50000	50000	50000	
Discount factor (@12%)	1	0.8929	0.7972	0.7118	0.6355	0.5674	
PV of Yearly Benefits	0	44642.9	39859.7	35589	31775.9	28371.3	
NPV of All Benefits		44642.9	84502.6	120092	151867	180239	180239
Initial Investmnet	(42500)	0	0	0	0	0	
Recurring Cost	0	(28500)	(28500)	(28500)	(28500)	(28500)	
	1	0.8929	0.7972	0.7118	0.6355	0.5674	
PV of Yearly Recurring Cost	(42500)	(25446)	(22720)	(20286)	(18112)	(16172)	
NPV of All Cost	(42500)	(67946)	(90666)	(110952)	(129064)	(145236)	(145236)
Overall NPV (NPV of All Benefits- NPV of All Cost)							35002.7
Overall ROI (Overall NPV/NPV of All Costs)							24.10
Break Even Analysis							
Yearly NPV Cash Flow (PV of Yearly Benefit- PV of Yearly Cost)	(42500)	19196	17140	15303	13664	12200	
Cumulative/Overall NPV cash flow	(42500)	(23304)	(6164)	9139	22803	35003	
Project Break -Even Occurs between years 2 and 3							
Use First Year of Positive Cash flow to calculate break even ratio = (15303-9139/15303)= 0.403							
Actual Break Even Occurs at 2.4 year =2+6164/15303) = 2.4							

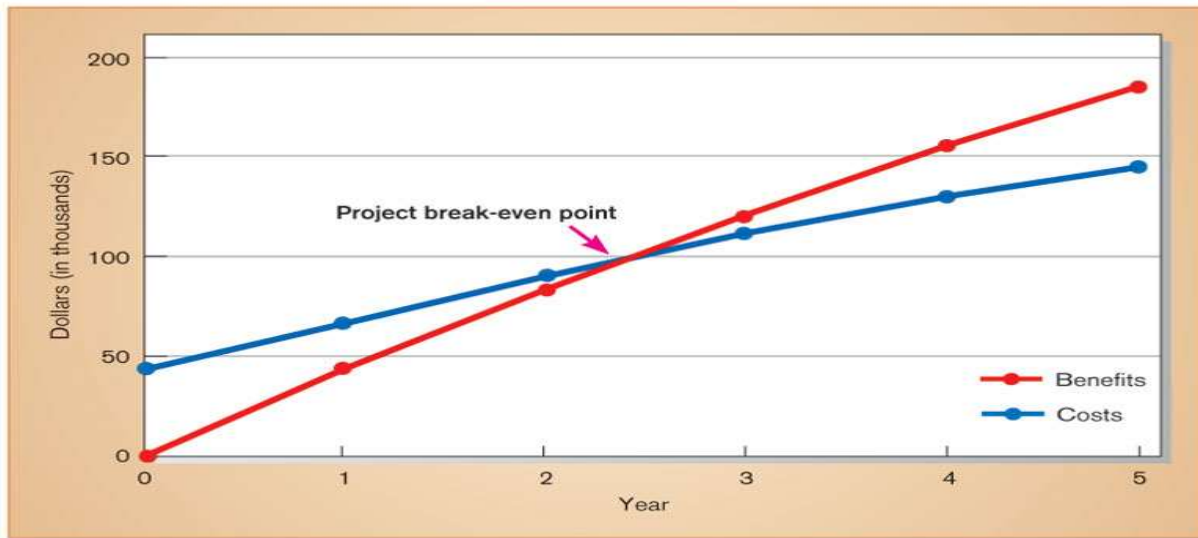
Break Even Analysis

$$\text{Break-Even Ratio} = \frac{\text{Yearly NPV Cash Flow} - \text{Overall NPV Cash Flow}}{\text{Yearly NPV Cash Flow}}$$

$$\text{Break - Even Ratio} = \frac{15,303 - 9139}{15,303} = .403$$

- worksheet shows that breakeven occurs between years 2 and 3. Because year 3 is the first in which the overall NPV cash flow figure is nonnegative
- Project breakeven occurs at approximately 2.4 years. A graphical representation of this analysis is shown in Figure

Figure 5-8 Break-even analysis for Customer Tracking System (Pine Valley Furniture)



Exercise1

- Q1) Assuming monetary benefits of an information system at \$50,000 per year, one-time costs of \$90,000, recurring costs of \$40,000 per year, and a three-year time horizon. Use a 10 percent discount rate and round the numbers to two decimal places,
 - calculate the net present value of these costs and benefits of an information system.
 - Also calculate the overall return on investment of the project and then present a break-even analysis.
 - At what point does break-even occur?

	Year of Project				
	Year 0	Year 1	Year 2	Year 3	TOTALS
Net monetary benefit	0	50,000	50,000	50,000	
Discount Rate (10%)	$=1/(1+.1)^0$ =1	$=1/(1+.1)^1$ =0.91	$=1/(1+.1)^2$ =0.83	$=1/(1+.1)^3$ =0.75	
PV of benefits	\$0.00	\$45,500.00	\$41,500	\$37,500	
NPV of all BENEFITS	\$0.00	\$45,500.00	\$87,000.00	\$124,500.00	\$124,500.00
One-time COSTS	-\$90,000.00				
Recurring Costs	\$0.00	-\$40,000	-\$40,000	-\$40,000	
Discount Rate (10%)	1	0.91	0.83	0.75	
PV of Recurring Costs	\$0.00	-\$36,400	-\$33,200	-\$30,000	
NPV of all COSTS	-\$90,000.00	-\$126,400.00	-\$159,600.00	\$189,600.00	-\$189,600.00
Overall NPV					<u><u>-\$65,100.00</u></u>
Overall ROI - (Overall NPV/ NPV of all COSTS)					<u><u>-0.34</u></u>
Break-Even Analysis					
Yearly NPV Cash Flow	-\$90,000	\$9,100.00	\$8,300	\$7,500	
Overall NPV Cash Flow	-\$90,000	-\$80,900.00	-\$72,600	-\$65,100	
Project break-even occurs	does not occur.				

Q2) Change the one-time costs in question 1 to \$10,000 and redo the analysis

	Year of Project				TOTALS
	Year 0	Year 1	Year 2	Year 3	
Net monetary benefit	0	50,000	50,000	50,000	
Discount Rate (10%)	1	0.91	0.83	0.75	
PV of benefits	\$0.00	\$45,500.00	\$41,500	\$37,500	
NPV of all BENEFITS	\$0.00	\$45,500.00	\$87,000.00	\$124,500	\$124,500.00
One-time COSTS	-\$10,000				
Recurring Costs	\$0.00	-\$40,000	-\$40,000	-\$40,000	
Discount Rate (10%)	1	0.91	0.83	0.75	
PV of Recurring Costs	\$0.00	-\$36,400	-\$33,200	-\$30,000	
NPV of all COSTS	-\$10,000	-\$46,400	-\$79,600	-\$109,600	-\$109,600
Overall NPV					<u><u>\$14,900</u></u>
Overall ROI - (Overall NPV/ NPV of all COSTS)					<u><u>0.14</u></u>
Break-Even Analysis					
Yearly NPV Cash Flow	-\$10,000	\$9,100	\$8,300	\$7,500	
Overall NPV Cash Flow	-\$10,000	-\$900	\$7,400	\$14,900	

Project break-even occurs between years 1 and 2
 Use first year of positive cash flow to calculate
 break-even fraction = $((8300 - 7400)/8300)$ 0.11
 Actual break-even occurred at 1.11 years

Exercise 2

- Assuming monetary benefits of an information system at \$85,000 per year, one-time costs of \$75,000, recurring costs of \$35,000 per year, a discount rate of 12 percent, and a five-year time horizon, calculate the net present value of these costs and benefits of an information system. Also calculate the overall return on investment of the project and then present a break-even analysis. At what point does breakeven occur?
 - Change the discount rate for Problem to 10 percent and redo the analysis.
 - Change the recurring costs in Problem to \$40,000 and redo the analysis.
 - Change the time horizon in Problem to three years and redo the analysis

Exercise 3

- Assume monetary benefits of an information system of \$40,000 the first year and increasing benefits of \$10,000 a year for the next five years (year 1 = \$50,000, year 2 = \$60,000, year 3 = \$70,000, year 4 = \$80,000, year 5 = \$90,000).
- Onetime development costs were \$80,000 and recurring costs were \$45,000 over the duration of the system's life. The discount rate for the company was 11 percent.
 - Using a six-year time horizon, calculate the net present value of these costs and benefits. Also calculate the overall return on investment and then present a break-even analysis. At what point does breakeven occur?
 - Change the discount rate for Problem to 12 percent and redo the analysis.
 - Change the recurring costs in Problem to \$40,000 and redo the analysis.

Accessing Technical Feasibility

- Assessing the organization's ability to construct the proposed system
- Takes into account various project risk factors
- The potential consequences of not assessing and managing risks can include the following:
 - Failure to attain expected benefits from the project,
 - Inaccurate project cost estimates,
 - Inaccurate project duration estimates,
 - Failure to achieve adequate system performance levels, and
 - Failure to adequately integrate the new system with existing hardware, software, or organizational procedures.
- **Project Risk Factors:**
 - Project size
 - Team size, organizational departments, project duration, programming effort
 - Project structure
 - New vs. renovated system, resulting organizational changes, management commitment, user perceptions
 - Development group
 - Familiarity with platform, software, development method, application area, development of similar systems
 - User group
 - Familiarity with IS development process, application area, use of similar systems

Accessing Technical Risks

- Risk can be managed on a project by:
 - Changing the project plan to avoid risky factors,
 - Assigning project team members to carefully manage the risky aspects,
 - Setting up monitoring methods to determine whether or not potential risk is, in fact, materializing.
- The four primary factors associated with the amount of technical risk on a given project are:
 - Project size,
 - Project structure,
 - The development group's experience with the application and technology area,
 - The user group's experience with systems development projects and the application area
- Four general rules emerged as technical risk assessments:
 - *Larger projects are riskier than smaller projects.*
 - *A system in which the requirements are easily obtained and highly structured will be less risky than one in which requirements are messy, ill structured, ill defined, or subject to the judgment of an individual.*
 - *The development of a system employing commonly used or standard technology will be less risky than one employing novel or nonstandard technology.*
 - *A project is less risky when the user group is familiar with the familiar with the systems development process and application area than if unfamiliar.*

Figure 5-9 Effects of degree of project structure, project size, and familiarity with application area on project implementation risk

		Low Structure	High Structure
High Familiarity with Technology or Application Area	Large Project	(1) Low risk (very susceptible to mismanagement)	(2) Low risk
	Small Project	(3) Very low risk (very susceptible to mismanagement)	(4) Very low risk
Low Familiarity with Technology or Application Area	Large Project	(5) Very high risk	(6) Medium risk
	Small Project	(7) High risk	(8) Medium-low risk

Source: Adapted from: Corporate Information Systems Strategy and Management: *The Challenges of Managing in a Network Economy*, 6th ed., by L. M. Applegate, R. D. Austin, and F. W. McFarlan. Copyright © 2003. Reprinted by permission of The McGraw-Hill Companies.

High technical familiarity mitigates risk due to project size and structure. Low familiarity increases risk.

Accessing Other Project Feasibility Concerns

• Operational

- The process of **assessing the degree** to which a proposed system **solves business problems or takes advantage of business opportunities**.
- Does the **proposed system solve problems or take advantage of opportunities**?
- The **likelihood** that the project **will attain its desired objectives**, called operational feasibility.
- Its purpose **is to gain an understanding of the degree to which the proposed system will likely solve the business problems or take advantage of the opportunities outlined in the System Service Request or project identification study**.
- For a project motivated from information systems planning, operational feasibility **includes justifying the project on the basis of being consistent with or necessary for accomplishing the information systems plan**.
- Systems that have substantial and widespread impact on an organization's structure or procedures are typically **riskier** projects to undertake. So this type of feasibility will **help the project to be out from such type of risk**.

• Scheduling

- The process of assessing the degree to which the **potential time frame and completion dates** for all major activities within a **project meet organizational deadlines and constraints** for affecting change
- Can the project time frame and completion dates meet organizational deadlines?
- The purpose of assessing schedule feasibility is, **to gain an understanding of the likelihood that all potential time frames and completion date schedules can be met** and that meeting these **dates will be sufficient** for dealing with the needs of the organization.
- Further, detailed activities may only be feasible **if resources are available when called for in the schedule**.
- While assessing schedule feasibility you should also evaluate **scheduling trade-offs**.
- For example, factors such as project team size, availability of key personnel, subcontracting or outsourcing activities, and changes in development environments may all be considered as having a possible impact on the eventual schedule.

Accessing Other Project Feasibility Concerns

- **Legal and Contractual**

- The process of **assessing potential legal and contractual ramifications due to the construction of a system.**
- What are legal and contractual ramifications of the proposed system development project?
- In this area, you need to gain an understanding of any **potential legal ramifications** due to the construction of the system.
- Possible considerations might include **copyright or nondisclosure infringements, labor laws, antitrust legislation** (which might limit the creation of systems to share data with other organizations), **foreign trade regulations** (e.g., some countries limit access to employee data by foreign corporations), and **financial reporting standards**, as well as **current or pending contractual obligations.**
- Contractual obligations may involve **ownership** of software used in joint ventures, **license agreements** for use of hardware or software, **nondisclosure agreements** with partners, or elements of a labor agreement (e.g., a union agreement may preclude certain compensation or work-monitoring capabilities a user may want in a system).

- **Political**

- The process of **evaluating how key stakeholders within the organization view the proposed system.**
- How do key stakeholders view the proposed system?
- Because an information system **may affect the distribution of information** within the organization, and **thus the distribution of power**, the construction of an information system can **have political ramifications.**
- Those **stakeholders not supporting the project** may take steps to block, disrupt, or change the **intended focus of the project.**

Building the Baseline Project Plan

- **Baseline Project Plan (BPP)** is a document intended primarily to guide the development team.
- Sections:
 - Introduction
 - System description
 - Feasibility assessment
 - Management issues
- Project Scope statement is part of the BPP introduction.
 - Sections:
 - Problem statement
 - Project objectives
 - Project description
 - Business benefits
 - Deliverables
 - Expected duration

Figure 5-10 Outline of a Baseline Project Plan

BASELINE PROJECT PLAN REPORT	
1.0 Introduction	
A. Project Overview—Provides an executive summary that specifies the project's scope, feasibility, justification, resource requirements, and schedules. Additionally, a brief statement of the problem, the environment in which the system is to be implemented, and constraints that affect the project are provided.	
B. Recommendation—Provides a summary of important findings from the planning process and recommendations for subsequent activities.	
2.0 System Description	
A. Alternatives—Provides a brief presentation of alternative system configurations.	
B. System Description—Provides a description of the selected configuration and a narrative of input information, tasks performed, and resultant information.	
3.0 Feasibility Assessment	
A. Economic Analysis—Provides an economic justification for the system using cost-benefit analysis.	
B. Technical Analysis—Provides a discussion of relevant technical risk factors and an overall risk rating of the project.	
C. Operational Analysis—Provides an analysis of how the proposed system solves business problems or takes advantage of business opportunities in addition to an assessment of how current day-to-day activities will be changed by the system.	
D. Legal and Contractual Analysis—Provides a description of any legal or contractual risks related to the project (e.g., copyright or nondisclosure issues, data capture or transferring, and so on).	
E. Political Analysis—Provides a description of how key stakeholders within the organization view the proposed system.	
F. Schedules, Time Line, and Resource Analysis—Provides a description of potential time frame and completion date scenarios using various resource allocation schemes.	
4.0 Management Issues	
A. Team Configuration and Management—Provides a description of the team member roles and reporting relationships.	
B. Communication Plan—Provides a description of the communication procedures to be followed by management, team members, and the customer.	
C. Project Standards and Procedures—Provides a description of how deliverables will be evaluated and accepted by the customer.	
D. Other Project-Specific Topics—Provides a description of any other relevant issues related to the project uncovered during planning.	

Baseline Project Plan (BPP) is a document intended primarily to guide the development team.

Sections:

- 1) Introduction**
- 2) System description**
- 3) Feasibility assessment**
- 4) Management issues**

Figure 5-11 Statement of project scope (Pine Valley Furniture)

Pine Valley Furniture Statement of Project Scope		Prepared by: Jim Woo Date: September 12, 2005
General Project Information		
Project Name:	Customer Tracking System	
Sponsor:	Jackie Judson, VP Marketing	
Project Manager:	Jim Woo	
Problem/Opportunity Statement:		
Sales growth has outpaced the Marketing department's ability to accurately track and forecast customer buying trends. An improved method for performing this process must be found in order to reach company objectives.		
Project Objectives:		
To enable the Marketing department to accurately track and forecast customer buying patterns in order to better serve customers with the best mix of products. This will also enable PVF to identify the proper application of production and material resources.		
Project Description:		
A new information system will be constructed that will collect all customer purchasing activity, support display and reporting of sales information, aggregate data, and show trends in order to assist marketing personnel in understanding dynamic market conditions. The project will follow PVF's systems development life cycle.		
Business Benefits:		
Improved understanding of customer buying patterns Improved utilization of marketing and sales personnel Improved utilization of production and materials		
Project Deliverables:		
Customer tracking system analysis and design Customer tracking system programs Customer tracking documentation Training procedures		
Estimated Project Duration:		
5 months		

Project Scope statement is part of the BPP introduction.

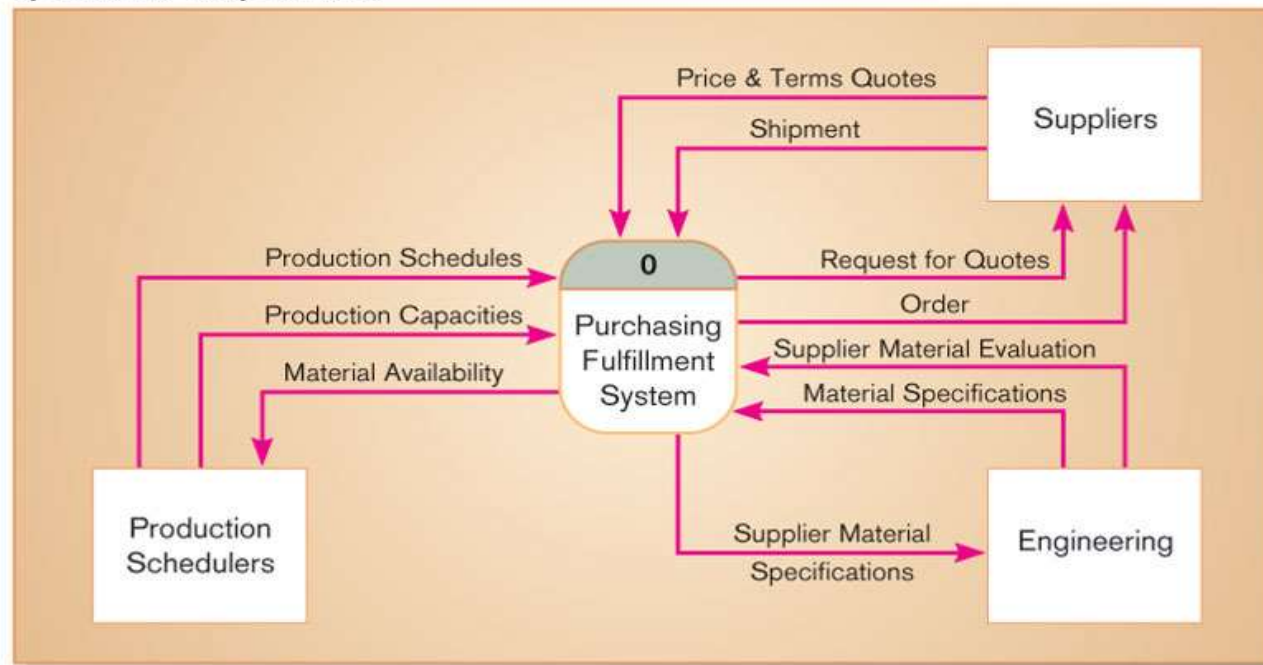
Sections:

- 1) Problem statement**
- 2) Project objectives**
- 3) Project description**
- 4) Business benefits**
- 5) Deliverables**
- 6) Expected duration**

Factors in Determining Scope

- Organizational units affected by new system
- Current systems that will interact with or change because of new system
- People who are affected by new system
- Range of potential system capabilities

Figure 5-12 Context-level data flow diagram showing project scope for Purchasing Fulfillment System (Pine Valley Furniture)



Building the Baseline Project Plan...contd...

- *System description* section outlines possible alternative solutions.
- *Feasibility assessment* section outlines issues related to project costs and benefits, technical difficulties, and other such concerns.
- *Management issues* section outlines a number of managerial concerns related to the project
- **Reviewing the Baseline Project Plan**
- **Structured Walkthroughs:** a peer-group review of any product created during the system development process
- Roles: coordinator, presenter, user, secretary, standard-bearer, maintenance oracle
 - Coordinator:
 - plans the meeting and facilitates a smooth meeting process.
 - may be the project leader or a lead analyst responsible for the current life cycle step.
 - Presenter:
 - describes the work product to the group.
 - User:
 - makes sure that the work product meets the needs of the project's customers
 - Secretary:
 - takes notes and records decisions or recommendations made by the group.
 - may be a clerk assigned to the project team or it may be one of the analysts on the team.
 - Standards bearer:
 - ensure that the work product adheres to organizational technical standards.
 - Many larger organizations have staff groups within the unit responsible for establishing standard procedures, methods, and documentation formats. These standards bearers validate the work so that it can be used by others in the development organization.
 - Maintenance oracle:
 - reviews the work product in terms of future maintenance activities. The goal is to make the system and its documentation easy to maintain.
- Can be applied to BPP, system specifications, logical and physical designs, program code, test procedures, manuals and documentation
- **Objectives**
 - **Assure conformity to organizational standards**
 - **All parties agree to continue with project**

Figure 5-13 Walkthrough review form (Pine Valley Furniture)

Pine Valley Furniture Walkthrough Review Form			
Session Coordinator: _____			
Project/Segment: _____			
Coordinator's Checklist:			
1. Confirmation with producer(s) that material is ready and stable: _____			
2. Issue invitations, assign responsibilities, distribute materials: ☐ Y ☐ N			
3. Set date, time, and location for meeting:			
Date: ____ / ____ / ____ Time: _____ A.M. / P.M. (circle one)			
Location: _____			
Responsibilities	Participants	Can Attend	Received Materials
Coordinator	_____	☐ Y ☐ N	☐ Y ☐ N
Presenter	_____	☐ Y ☐ N	☐ Y ☐ N
User	_____	☐ Y ☐ N	☐ Y ☐ N
Secretary	_____	☐ Y ☐ N	☐ Y ☐ N
Standards	_____	☐ Y ☐ N	☐ Y ☐ N
Maintenance	_____	☐ Y ☐ N	☐ Y ☐ N
Agenda:			
____ 1. All participants agree to follow PVF's Rules of a Walkthrough			
____ 2. New material: walkthrough of all material			
____ 3. Old material: item-by-item checkoff of previous action list			
____ 4. Creation of new action list (contribution by each participant)			
____ 5. Group decision (see below)			
____ 6. Deliver copy of this form to the project control manager			
Group Decision:			
____ Accept product as-is			
____ Revise (no further walkthrough)			
____ Review and schedule another walkthrough			
Signatures	_____	_____	_____

Structured
walkthrough form